

Credit Cards and Retail Firms: Historical Evidence from the U.S.

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Overview

This study examines the small firm not as a *borrower*, but as a *lender*—a role that small businesses frequently played in the U.S. before the rise of bank credit cards, and one they continue to play worldwide

Why it matters for policy:

- ▶ Merchants pay 2–4% interchange fees on every card transaction
- ▶ Regulators worldwide debating whether to cap these fees (EU, UK, US Credit Card Competition Act)
- ▶ But, bank cards replaced even more expensive in-house credit operations

Main finding: Bank credit cards allowed small firms to *outsource* the credit supply function, *reducing* their costs net of interchange fees

Roadmap

1. **Historical Context:** How store credit worked and why banks replaced it
2. **Aggregate Trends:** Banks displaced stores as credit suppliers since 1958
3. **Causal Evidence:** 1978 Supreme Court shock to bank card supply
 - ▶ Store lending fell, firm profits and entry increased
4. **Mechanisms:** Cost reduction (labor, risk) dominates markup compression
5. **Policy Implications:** Interchange fees vs. in-house credit costs

Historical Context

“Most Americans had a dozen or more different forms of credit...an account at the neighborhood pharmacy, the neighborhood grocery store, and a half-dozen other local stores”

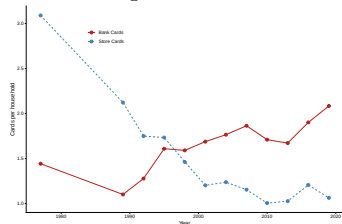
“It was the smaller merchants who first came around...His accounts receivable were dragging him under...I looked at the size of his accounts: \$4.58. \$12.82. He was sending out monthly bills.”

“The bank would guarantee him payment—within days instead of months—and would take over the role of collecting from the customers... [while] taking a 6 percent cut.”

—Bank of America salesman, 1958 (Nocera 2013)

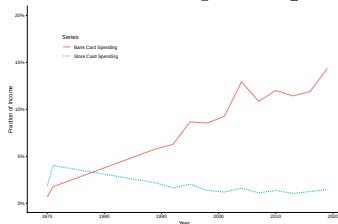
Bank Cards Displaced Store Credit

Cards per household



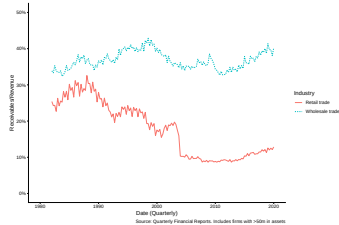
Source: SCF

Household spending



Source: SCF

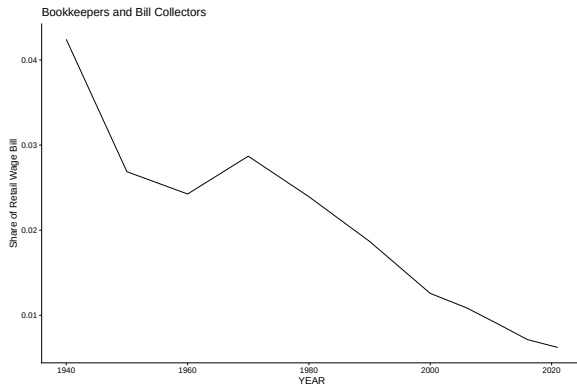
Firm receivables



Source: US Census QFR (assets $\geq$ \$50M)

Firms Saved on Credit-Related Labor

Retail firms once spent 4% of their wage bill on bookkeepers and bill collectors



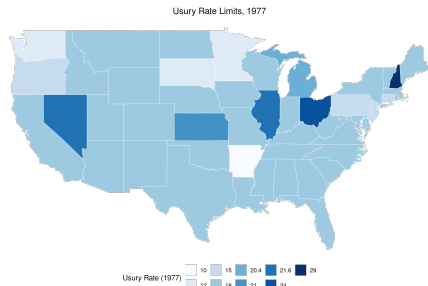
Source: US Census.

Identification: 1978 *Marquette* Decision

- ▶ Supreme Court ruled banks can lend across state lines at *home state's* usury rate
- ▶ States with usury $< 18\%$ in 1977 saw sudden increase in bank card supply from out-of-state banks
- ▶ 9 “treated” states (37M people, 16% of nation)
- ▶ Revolving credit regulated separately—no effect on mortgages, business loans
- ▶ I do *not* use the subsequent endogenous deregulation decisions: only the unexpected Supreme Court shock for identification

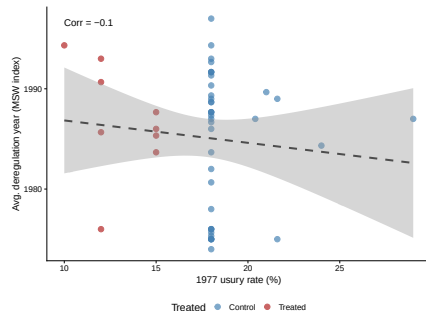
Support for Identification

No geographic pattern



Source: Cost of Personal Borrowing, 1977

Orthogonal to other deregulation

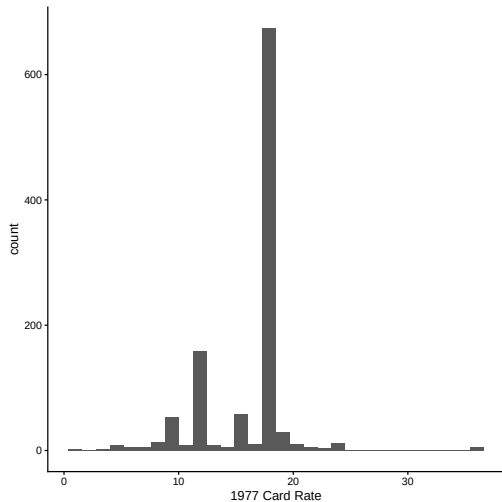


Corr = -0.10 with Mian-Sufi-Verner deregulation index

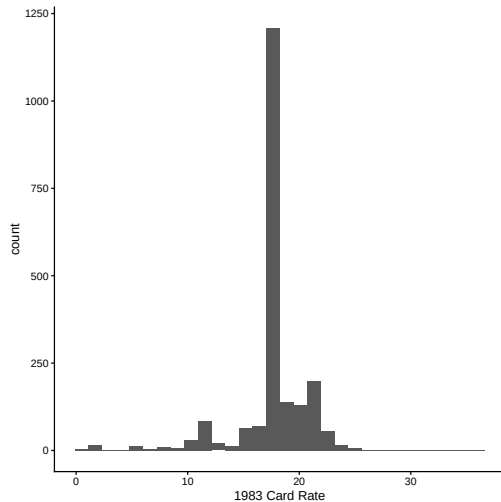
Treatment balance: no significant differences in wages, employment, GDP, or retail share between treated and control states in 1977

Usury Limits Were Binding

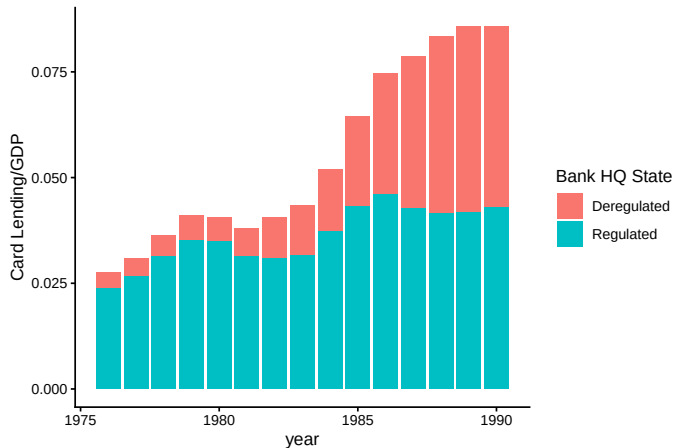
1977: Bunching at limits



1983: Limits no longer bind



Card Lending Grew from Deregulated-State Banks



Card lending / GDP by bank HQ state usury regime. Credit card lending tripled 1976–1990; almost all growth from banks headquartered in deregulated states.

New Data on Bank Card Acceptance

MASTER CHARGE

One Bill - One Check
Shopping convenience at
Thousands of Merchants in
this area.
Look for the Master Charge
Sign at the Merchants
Shown Below



BANKS

WEST SIDE BANK

MainAv&JacksonScrn--346-8767

ART GALLERIES & DEALERS

GALERIE des BEAUX ARTS

352AdmsAvScrn--343-8591

AUTOMOBILE AIR CONDITIONING EQUIPMENT

KAPLAN'S SERV AUTO RADIATOR CO

810WyoAvScrn--344-3914

AUTOMOBILE BODY REPAIRING & PAINTING

DERENICK'S REPR & BODY

SHOP 460 N MainTay-----562-1260

KAPLAN'S SERV AUTO RADIATOR CO

810WyoAvScrn--344-3914

TIBERI'S BODY SHOP

Rr708 N MainAvScrn--344-5118

Credit Card & Other Credit Plans

PITTSBURGH NATIONAL BANKAMERICARD



CHARGE CARD

Lets You Do Your
Shopping At Merchants
Most Convenient To You.

"FOR INFORMATION CALL"

BANKAMERICARD MERCHANTS

OAKMONT

Four The Girls Shop

434AlleghenyRiverBlvd
Oakmont--828-4406

Verona Auto Parts Co

827AlleghenyAvOakmont--828-8500

PENN HILLS

Associated Hardware

EastHillsShopgCtr--731-4845

Penn Hills Auto Parts

119PennoakDrPittsbrgh--242-8626

Penn Hills Electronic Service

1924UniversalRdPennHills--793-3005

Russ Plumbing

11752FrankstownRd--242-0111

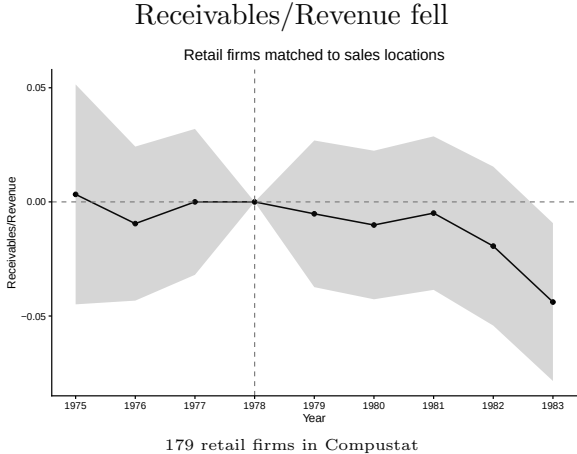
STANTON HEIGHTS

Associated Hardware

StantonHtsShopgCtr--661-2654

- ▶ U.S. Telephone Directory Collection, Library of Congress
- ▶ 1.97M firm-years screened in Dun & Bradstreet
- ▶ 141K covered by phonebooks across 418 municipalities
- ▶ 510 card-accepting merchant-years manually transcribed and matched to D&B
- ▶ Final sample: Compustat firms matched to D&B sales locations, with sales from D&B and card acceptance from phonebooks (2,956 firms, 283 markets, 1970-1977)

Store Lending Fell, Profits Rose



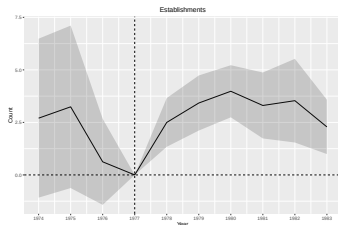
Dependent Variable:	Net Income / Revenue
Model:	(1)
<i>Variables</i>	
Treated $\times$ Post	0.0091* (0.0054)
<i>Fixed-effects</i>	
Year FE	Yes
Firm FE	Yes
<i>Fit statistics</i>	
Observations	2,000
R ²	0.21338
Within R ²	0.00058

Clustered (Firm FE) standard-errors in parentheses
Signif. Codes: ***: 0.01, **: 0.05, *: 0.1
DV: net income / revenue (post-1978 level shift)
At 5%, I reject that profits fell by more than 15bp

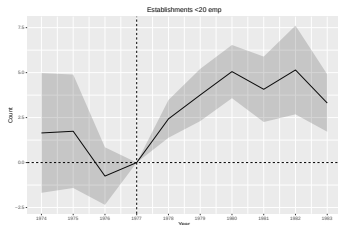
Net Entry: Triple-Difference by Industry

- **Why net entry?** Accounting profits are a flawed proxy for economic profits and are observed only for the largest firms (Compustat); net entry captures the competitive response of the whole market.
- **Why triple-difference?** Entrepreneurs also use credit cards to *fund* their businesses. The triple-difference isolates the *payment* channel: clothing retail had the highest card payment intensity, independent of card use by owners as borrowers.

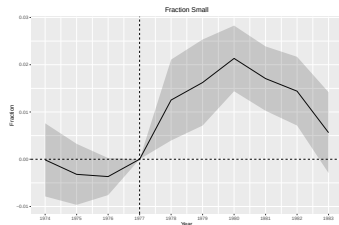
Compare to other retail sub-industries:



(a) Establishments



(b) Small establishments



(c) Fraction small

4% increase in establishments; entrants are disproportionately small.

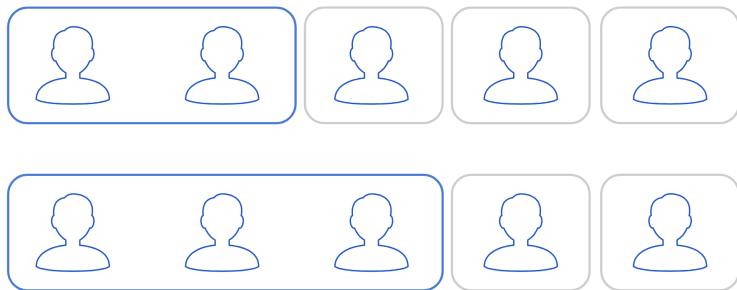
Which Firms Adopt Bank Cards?

Dependent Variable: Model:	Card acceptance	
	(1)	(2)
<i>Variables</i>		
log(Sales)	0.7112*** (0.1500)	0.7243*** (0.1505)
Single-establishment	0.6874*** (0.2410)	0.7132*** (0.2308)
Firm age		-0.0075 (0.0071)
<i>Fixed-effects</i>		
Industry $\times$ City FE	Yes	Yes
<i>Fit statistics</i>		
Observations	1,940	1,940
Squared Correlation	0.23179	0.23384
Pseudo R ²	0.24409	0.24501
BIC	1,522.3	1,528.6
<i>Clustered (industry) standard-errors in parentheses</i>		
<i>Signif. Codes: ***: 0.01, **: 0.05, *: 0.1</i>		

- ▶ Geographically concentrated, high-sales firms adopt first
- ▶ Consistent with cost reduction (undiversified firms benefit most)
- ▶ Inconsistent with demand-steering (low-share firms benefit most from stealing business)

The Model in One Picture

Each face is a **retailer**; all retailers compete for customers, but *harder* against rivals that accept the **same payment method** — the grouped outlines below.



- ▶ Boxes = card-acceptance **nests**; stores inside a nest are closer substitutes.
- ▶ **Marquette** reshapes the nesting structure: as bank cards spread, more stores join the accepting nest (top row → bottom row).
- ▶ Calibration asks: **how much does the nest matter for where consumers shop?**
That is what σ measures.

Calibrating to the *Marquette* Experiment

Nested logit: $u_{ij} = \delta_j + \varsigma_{g(j)} + (1 - \sigma)\varepsilon_{ij}$, where $\delta_j = \alpha p_j + \xi_j$

- ▶ σ = fraction of taste dispersion correlated *within* card-acceptance groups
- ▶ $\sigma \rightarrow 1$: customers choose only on payment method, not what the store sells
- ▶ $\sigma = 0$: payment method irrelevant; choice is on price and idiosyncratic taste

How I calibrate: I *simulate Marquette inside the model* — raising consumer bank-card usage and merchant acceptance to match the data — and pick (σ, ζ) so the model reproduces the **actual diff-in-diff effects of Marquette**, estimated across my 252 retail markets:

1. the average sales response of treated firms (*Treated* $\times$ *Post*), and
2. the *extra* sales response of firms already accepting bank cards (*Treated* $\times$ *Post* $\times$ *Accept*).

α (price sensitivity) is calibrated from the literature; only σ and ζ (the cost change) are matched. Two moments, two parameters — exactly identified.

Calibrated Model: Cost Reduction vs. Competition

Question: Does accepting bank cards mainly (a) reduce costs or (b) steal customers?

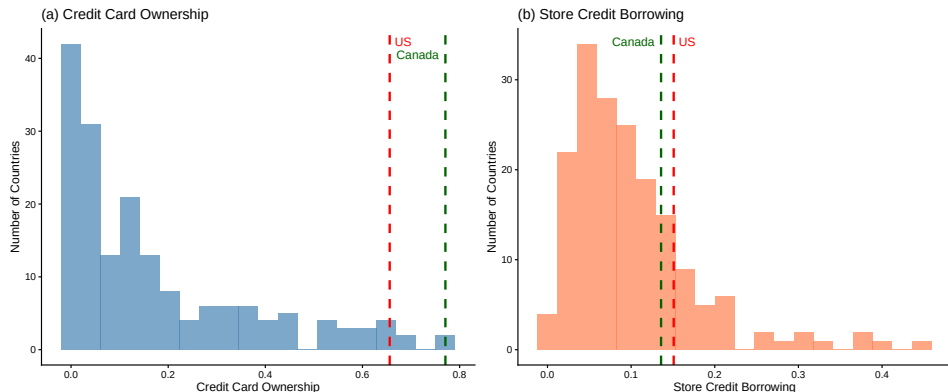
Calibrated Parameters	
Parameter	Value
α (price sensitivity)	-4
σ (card differentiation)	0.017 (0.025)
ζ (cost change)	-0.105 (0.055)
Implied cost reduction	3.9%

Delta-method s.e. in parentheses

Decomposition		
	Cost	Markup
Baseline $\rightarrow$ Marquette	-80bp	-5bp
Cost only	-80bp	+1bp
Competition only	-16bp	-6bp

First decomposition of retail demand into product taste vs. payment-method taste: point estimate $\sigma \approx 0.02$ is modest but economically real; 95% CI bounds it above at ≈ 0.07 .

International Context



Many developing countries still rely heavily on store credit. The U.S. historical experience suggests that enabling bank-provided consumer credit can reduce costs for small merchants.

Conclusion and Policy Implications

Summary of findings:

- ▶ Bank cards displaced store credit, reducing merchants' labor and risk costs
- ▶ Net effect on retail industry: +4% entry, +2% profits, better recession survival
- ▶ Cost reduction ($\sim 4\%$) dominates markup compression (5bp)

Implications for interchange fee regulation:

- ▶ Interchange fees replaced even higher in-house credit costs
- ▶ Small merchants benefit most from outsourcing credit to banks
- ▶ Capping fees may force merchants back toward costly in-house credit
- ▶ Historical evidence: the *net* effect of card networks on merchants was positive

Backup: Sensitivity to α

α	σ	ζ	Cost reduction
-2.000	0.0157	-0.2569	-0.0951
-3.000	0.0167	-0.1499	-0.0555
-4.000	0.0173	-0.1048	-0.0388
-5.000	0.0176	-0.0802	-0.0297
-6.000	0.0109	-0.0531	-0.0197
-0.258	0.0099	-2.3915	-0.8849

Cost reduction ranges from about 2% ($\alpha = -6$) to 9.5% ($\alpha = -2$). σ is stable across specifications (0.01–0.02).

Backup: Card Acceptance by Industry

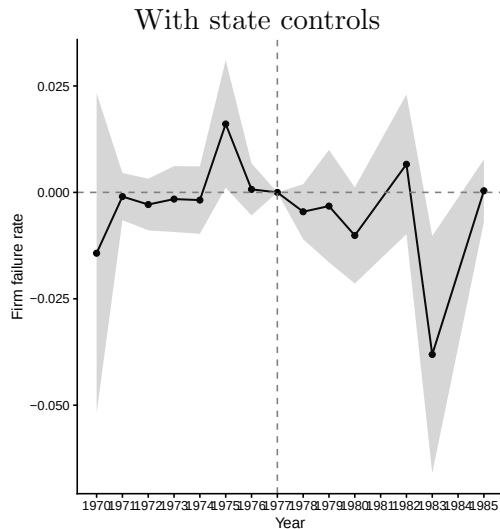
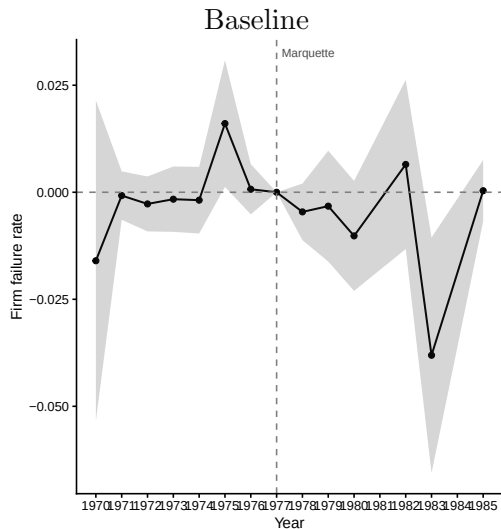
industry	acceptance	n	firms_accepting
Clothing	1.5%	4513	67
Furnishing	0.7%	4686	34
Misc.	0.6%	11982	68
Auto	0.5%	10154	46
Department Stores	0.3%	2332	7
Hardware	0.3%	3112	9
Restaurants	0.1%	12794	17
Food	0.0%	8247	2

Backup: Recession Survival

Dependent Variable: Model:	Exit rate			
	(1)	(2)	(3)	(4)
<i>Variables</i>				
Mean log(Sales)	-0.0092*** (0.0004)	-0.0127*** (0.0008)	-0.0107*** (0.0005)	-0.0155*** (0.0008)
Treated × Single-estab.	-0.0006 (0.0027)	-0.0016 (0.0026)	-0.0020 (0.0015)	-0.0041** (0.0019)
Single-estab. × Recession	0.0459*** (0.0039)	0.0514*** (0.0046)	0.0305*** (0.0033)	0.0159*** (0.0022)
Treated × Recession × Single-estab.	-0.0159* (0.0081)	-0.0204** (0.0099)	0.0003 (0.0113)	0.0072 (0.0084)
<i>Fixed-effects</i>				
State × Industry × Year FE	Yes	Yes	Yes	Yes
<i>Fit statistics</i>				
Observations	73,036	27,252	73,036	27,252
R ²	0.99726	0.99720	0.99718	0.99704
Within R ²	0.11503	0.17114	0.08924	0.12386

Clustered (STATE) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Backup: DiD with Controls



Results robust to controlling for baseline state wages and employment interacted with year FE.